

HPV-mediated cervical carcinogenesis: concepts and clinical implications.

Persistent infection with a high-risk human papillomavirus (hrHPV) is generally accepted as a necessary cause of cervical cancer. However, cervical cancer is a rare complication of an hrHPV infection since most such infections are transient, not even giving rise to cervical lesions. On average, it takes 12-15 years before a persistent hrHPV infection may ultimately, via consecutive premalignant stages (ie CIN lesions), lead to an overt cervical carcinoma. This argues that HPV-induced cervical carcinogenesis is multi-step in nature. In this review, the data from hrHPV-mediated in vitro transformation studies and those obtained from analysis of clinical specimens have been merged into a cervical cancer progression model. According to this model, a crucial decision maker in the early stages following infection involves individual susceptibility for certain HPV types depending on the genetic make-up of immune surveillance determinants. Once a CIN lesion has developed, altered transcriptional regulation of the viral E6/E7 oncogenes, resulting in genomic instability and distinguishing the process of cell transformation from a productive viral infection, probably provides the subsequent important step towards malignancy. The additional (epi)genetic alterations that subsequently accumulate in high-grade CIN lesions may result in overt malignancy via immortality and growth conditions that gradually become less sensitive to growth-modulating influences mediated by cytokines and cell-cell and cell-matrix adhesions. The potential implications of hrHPV testing and some other biomarkers deduced from this model for cervical screening and the clinical management of CIN disease are also discussed. Copyright 2006 Pathological Society of Great Britain and Ireland. Published by John Wiley & Sons, Ltd.